

Gravity: Summer Research Lecture

Newtonian Gravity:

$$F = \frac{Gm_1m_2}{r^2}$$

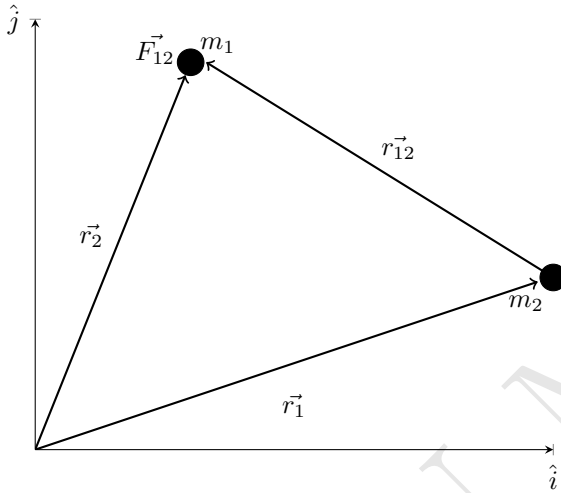
$$G = 6.67 \times 10^{-11} \frac{Nm^2}{kg^2}$$

is the magnitude of the force between two bodies of mass m_1 and m_2

the vector form is

$$\vec{F}_{12} = -\frac{Gm_1m_2}{r_{12}^2} \hat{r}_{12}$$

where $\vec{r}_{12} = \vec{r}_1 - \vec{r}_2$, $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$, and $\vec{F}_{12}$ is the force on m_1 from m_2 . r_{12} points from m_2 to m_1 and the negative sign in our $\vec{F}_{12}$ equation makes it attractive as gravity is an attractive force.



the force on m_1 (object of interest) from m_2 (source). m_2 sources the gravitational field that m_1 experiences

$$\vec{F}_{on1} = m_1 \vec{\mathfrak{G}}$$

where $\mathfrak{G}$ is the gravitational field

When there are many particles we have

$$\vec{F}_{ij} = \vec{F}_{i1} + \vec{F}_{i2} + \dots + \vec{F}_{iN}$$

excluding $i = j$ (you cannot exert a force on yourself)

$$= \sum_{j \neq i} \vec{F}_{ij}$$

Like so many things, there are direct relationships between forces and potential energy

$$\vec{F} = -\vec{\nabla} P.E.$$

When force is conservative (like in potential energy functions) you can take the negative gradient of the potential energy and it will equal the force of the system. In cartesian:

$$\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Gravitational Potential:

$$\Phi = \frac{U}{m}$$

The gravitational potential is represented by Φ and is equal to the potential energy over the mass. Similar to electrostatics where electric potential is equal to the potential per charge.

Potential energy of a point source $U_{pointsource}$ in terms of gravity G , the masses of the two objects in question m_1, m_2 , and the distance between them r

$$U_{point\ source} = -\frac{Gm_1m_2}{r}$$

Note that the point source potential is over r and not r^2 . This is due to the potential energy being derived from the work done against the gravitational force and not calculating the force itself. Force is equal to negative gradient of potential energy (which is just the vector derivations) so it's natural that the potential energy is going to be an integrand of the force.

These two can be combined to get the gravitational potential Φ for a point source *this one is really important* (reminder that since gravity is always an attractive force, it is always negative in our equations)

$$\Phi = -\frac{Gm_{source}}{r} = \frac{U}{m_{interest}}$$

Now, let's get into acceleration calculations to tie it all in together.

$$\frac{\vec{F}}{m} = -\vec{\nabla} \frac{U}{m} \rightarrow \vec{a} = -\vec{\nabla} \Phi$$

What is this? How did we get here? Let's break it down.

$$\vec{F} = ma \rightarrow \vec{a} = \frac{\vec{F}}{m}$$

now is a good time to recall that $\vec{F} = -\vec{\nabla} U$, $\Phi = \frac{U}{m}$, and U is the potential energy therefore:

$$\vec{a} = -\vec{\nabla} \left[-\frac{Gm}{r} \right] \rightarrow \vec{a} = -\frac{1}{m} \vec{\nabla} U \rightarrow \vec{a} = -\vec{\nabla} \Phi$$

Question: Since this has been broken down to the absolute basics, I am going to test your general knowledge.

What are the units of Φ and $\vec{\nabla} \Phi$?

For a system of N-bodies:

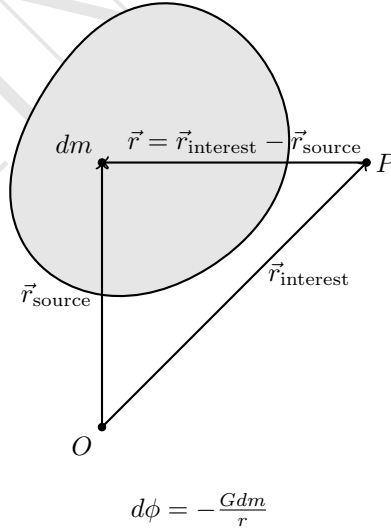
$$\begin{aligned}\vec{a} &= -\vec{\nabla} \sum_{j \neq i} \Phi_j \\ &= -\vec{\nabla} \sum_{j \neq i} \left[-\frac{Gm_j}{r_{ij}} \right]\end{aligned}$$

note that $\vec{\nabla}$ is evaluated at the position of interest, $\vec{r}_i$

Question: Compute $-\vec{\nabla}[-\frac{Gm}{r}]$ for $\vec{\nabla} = \hat{r} \frac{\partial}{\partial r}$ to show that $-\vec{\nabla}\Phi$ recovers the acceleration expression you are familiar with. Do the same but for $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ to get the expression in x, y, z coordinates.

For globular cluster modeling, we model the globular cluster stars as point particles, i.e $\Phi_{G.C} = \sum_j \Phi_j$, $\Phi_i = -[\frac{Gm_i}{r_i}]$, but the galaxy is modeled as a continuous mass distribution, which includes both the luminous mass of the galaxy and the dark matter halo mass.

Continuous mass distribution



From our jelly bean/blob, we break it up into little segments. dm_i and dm_j where dm is a mass element. The potential is then the sum over all contributions from all mass elements. The potential for the i^{th} element

is equal to:

$$d\phi_i = -\frac{Gdm_i}{|\vec{r} - \vec{r}_i|}$$

$|\vec{r} - \vec{r}_i|$ shows the magnitude of the distance from origin to point of interest (P) minus the distance from origin to our mass element.

mass elements all have mass dm and volume dv_i , from $\rho = \frac{m}{v}$ then $dm = \rho dv$ giving us density adding in $\rho(\vec{r}_i)$ is the density at source location $\vec{r}_i$

$$d\phi_i = -\frac{G\rho(\vec{r}_i)}{|\vec{r} - \vec{r}_i|} dv_i$$

then doing that for all of our mass elements, we sum them up giving

$$\Phi = -\sum_i \frac{G\rho(\vec{r}_i)}{|\vec{r} - \vec{r}_i|} dv_i \quad \text{lim as } dm \rightarrow 0$$

$$\Phi = \int d\phi = -G \int \rho \frac{(\vec{r}')}{|\vec{r} - \vec{r}'|} dv'$$

Giving, in total, the potential for a continuous source. Given that density can change, this is a function of where you are at within the jelly bean. Point of interest does not change and the integral does not depend on r , but in being over r' , you sum over all of the points within the jelly bean and each point is a vector from the origin.

Why does this all matter? N-body modeling consists of several components. The globular cluster (assume a halo cluster) orbits around the galaxy. The galaxy consists of 3 parts: the bulge, the disk, and the halo, all made up of stars (individual little pieces). We cannot model every single star so we give a galactic potential. The galactic potential is described as a continuous mass distribution as it is impossible to model every single star. N-body calculations then model the interactions between individual cluster stars (still a lot of stars and also still not all of the stars bc that would take entirely too long) and the different parts of the galaxy.

$$\Phi_{galactic} = \Phi_{disk} + \Phi_{bulge} + \Phi_{halo}$$

The galaxy is held within a spherical distribution of matter we cannot see, called the dark matter halo. Each potential is modeled as a continuous mass distribution coming together to form one giant continuous mass distribution. The acceleration on an individual star is then

$$\vec{a}_i = -\vec{\nabla}[\Phi_{G.C} + \Phi_{galactic}]$$

our globular cluster (G.C) is all of our point particles. The galactic of course being our continuous mass distribution of all of our galactic potentials.

Collisional vs Collisionless simulations

Collisional:

- models two body close encounters

- gravitational force is very strong as they are very close together
- individual particle-particle close encounters are important

Collisionless

- This is what we care about more as focus is on tidal stripping rather than internal mechanics of G.C.
- larger groupings negate need for close encounters anyway
- graininess of individual particles is an artifact of the simulation, not the physics
 - single particle can and may represent many particles
- point source potential for collisionless is $\phi = -\frac{Gm}{\sqrt{r^2 + \epsilon^2}}$ where ϵ is the softening length which is a fixed artificial distance keeping things from getting too close to one another.

bonus tidal forces

take the earth and the moon. there is a center to center distance (r), and two points (a, b) on either side of the moon. the gravitational strength at point a is stronger than the gravitational force at point b as point a is on the close side of the moon. While it may not be a lot, the two sides of the moon exert different levels of force on the earth. This accounts for the tides we experience. Everyone knows the earth exerts a force on the moon (thats why it's there at all, a nice little forced friend) but the moon also exerts a force back on us. We are a planet full of water, water as a liquid is much happier to go with the flow in comparison to the solid rock of the rest of the planet. Water will deform to whatever environment it's put it, glass, puddle, cube, sink, cliff, gravitational pull of giant space rock, etc.

the earth is covered in this easily deformable substance that will happily move along with the gravitational differential. The water puffs up with the orbit of the moon giving us tides.

comparing the force difference.

$$F_A - F_B = -GmM\left(\frac{1}{(r - \frac{\Delta r}{2})^2} - \frac{1}{(r + \frac{\Delta r}{2})^2}\right)$$

r is the center to center distance and Δr is the difference between points a and b

equating this to derivatives leads us back to their old friend the taylor series!

$$f(x + \Delta x) = f(x) + \frac{\partial f}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 f}{\partial x^2} + \dots$$

but we only need the first order expansion

$$\begin{aligned} f\left(r + \frac{\Delta r}{2}\right) &= \frac{1}{(r \pm \frac{\Delta r}{2})^2} \rightarrow \frac{1}{r^2} \pm \left[\frac{d}{dr} \frac{1}{r^2}\right] \frac{\Delta r}{2} \\ &= \frac{1}{r^2} \pm \left[-\frac{1}{r^3}\right] \frac{\Delta r}{2} \rightarrow \frac{1}{r^2} \pm \frac{1}{r^3} \frac{\Delta r}{2} \end{aligned}$$

Question: double check that the units are still correct for the final taylor expansion of $F_A - F_B$

continuing the calculation

$$F_A - F_B = -GmM\left(\frac{1}{r^2} - \frac{1}{r^3}\frac{\Delta r}{2} - \frac{1}{r^2} - \frac{1}{r^3}\frac{\Delta r}{2}\right)$$

$$F_{tide} = \frac{GmM}{r^3}\Delta r$$

This is the tidal field, the force differential between sides a and b that accounts for the tidal force. This tidal force is due to a gradient in forces across the moon (or across the earth depending on how you look at it)

$$\frac{1}{r^3}\Delta r$$

Lets try a calculation. Try $\Phi = \frac{-GM}{r}$

$$F_{tidal} = m\left[\frac{d^2}{dr^2}\Phi\right]\Delta r$$

(you'll notice this is not the gradient, this is the second derivative)

$$= m\frac{d}{dr}\left[\frac{GM}{r^2}\right]\Delta r$$

take the derivative again

$$m\frac{GM}{r^3}\Delta r$$

we have recovered F_{tidal} by taking the second derivative of the potential. When dealing with gravity, you will encounter the tidal tensor. This is the second derivative of the potential and has 6 components. With 3 coordinates you must take the second derivative of each coordinate and all of the cross terms.

tidal review: tidal field is consequence of force gradient across the moon, to get this we have to look at the differences between those forces. To the first order of the taylor series expansion we found the difference to be $\frac{1}{r^3}\Delta r$. we can also get this by taking the second derivative of the gravitational potential while including Δr .

The galactic Tidal tensor $\Phi_{ij} = \Phi_{xx}\Phi_{xy}\Phi_{xz}\Phi_{yy}\Phi_{yz}\Phi_{zz}$. say $\Phi_{xx} = \frac{d^2}{dx^2}\Phi$ and $\Phi_{xy} = \frac{d}{dx}\Phi$ (establishing subscripts are derivatives). There are only 6 unique components calculated through second derivatives of the potential.

The globular cluster is a ball of stars, a galaxy within a dark matter halo, and a force differential across the

globular cluster. This creates a tidal field. The consequence of this is stars being stripped from the cluster forming tidal tails.

The tidal tensor gives information about the force differential which gives information about the tidal field, which in turn gives information about the particles will move within the tidal field.

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